**Question 2**

**A1** (Measure) p.d. and find a force (could be in a table or graph). Change p.d. and repeat. 1
i.e. correct procedure. Do not give for only identifying the variables.
This mark can be scored if the method is unworkable.

**A2** Measurement of p.d. – voltmeter connected in parallel with capacitor or power supply on diagram or voltage read from power supply in text. Wrong diag loses this mark. 1

**A3** Workable electrical arrangement: d.c. power supply on diag or in text. 1
Ignore Voltmeter in series. Ammeter in parallel loses this mark.
Do not allow discharge of circuit while measuring force.
If diagram wrong this loses this mark.

**A4** Method of changing the p.d. (diag or text). 1
Variable power supply or potential divider circuit. Do no allow variable resistor in series.
Do not allow changing distance.

**B1** Measure the force/(mass) with tpb or force with newtonmeter. 1
Or measure force from a mass/lever/pulley system.

**B2** Workable mechanical arrangement to measure force on diag. 1
Allow workable loading/lever/pulley system.

**B3** **Plates** close together/large surface area or HT/EHT/hight voltage to 1
make force large/measurable.

**C** Any one safety precaution with reason. 1
e.g. wear insulating/rubber gloves: earth/insulate one plate wrt top pan balance or newtonmeter; discharge before touching plates to avoid/prevent shock/short circuit.
Do not allow not touching plates (precaution needed).

**D1/2** Any further good design features. 2
Some of these might be:
Two fixed points shown on the diagram (appropriate clamps or supports).
Distance between the plates should be kept constant.
Method of achieving constant separation of plates (spacers or measurement; no insulators or dielectrics)/adjust newtonmeter position to give same separation.
Force = mg
Force is the change in readings.
Calibrate springs to give force reading.
Approx. p.d (greater than 100 V)/distance (less than 1 cm)/Area (greater than 400 cm2)

**10 marks in total.**

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